

Module 2

Internal Combustion Engines: Air standard cycles – Otto and Diesel cycles, working of two stroke and four stroke Petrol and Diesel engines, Carburetted and MPFI engines, fuel pump, fuel injector, ignition system, cooling system, lubricating system.

Refrigeration & Airconditioning: Introduction to refrigeration and air -conditioning, Rating of refrigeration machines, Coefficient of performance, Simple refrigeration vapour compression cycle, summer and winter air conditioning.

Air standard cycles

- Air standard cycle is an idealised cycle in which air is taken as the working fluid
- Air standard cycles can be used for comparing the performance of real internal combustion engine

Air Standard Efficiency

- A hypothetical efficiency calculated from the ideal air standard cycle, in order to compare the efficiency of various cycles is called air standard efficiency or ideal thermal efficiency

Ideal thermal efficiency of an air standard cycle may be defined as the ratio of net work done to the heat supplied during the cycle

- The fraction of the heat input that is converted to net work output is a measure of the performance of a heat engine and is called the **thermal efficiency (η_{th})**

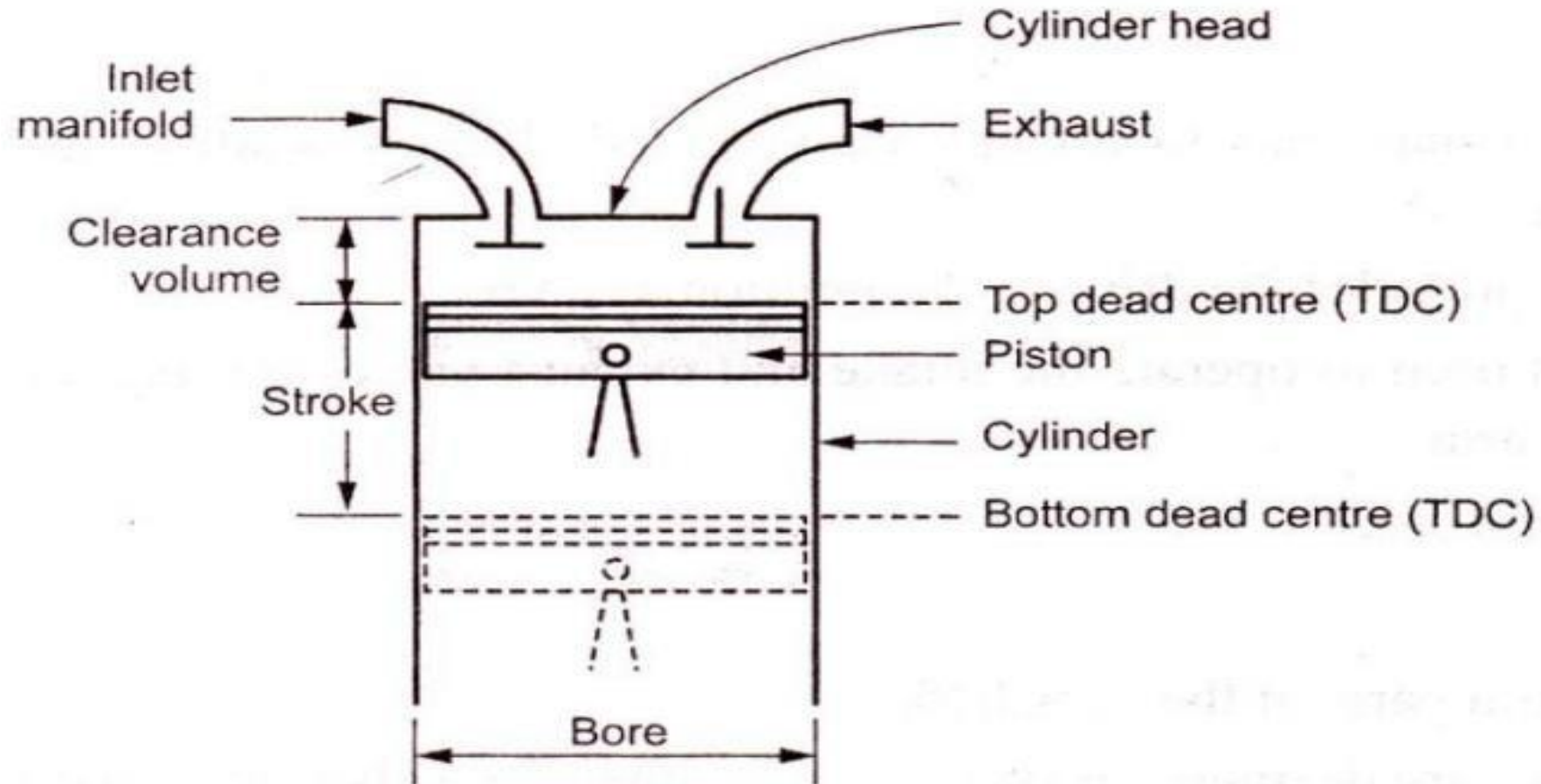
$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{th} = \frac{W_{net,out}}{Q_{in}}$$

$$\eta_{th} = 1 - \frac{Q_{out}}{Q_{in}}$$

TERMINOLOGIES RELATED TO RECIPROCATING ENGINE

IC ENGINE TERMINOLOGY



- **Bore** the internal diameter of the cylinder of the engine is called **bore**

- **Top dead centre(TDC):** Extreme position of the piston at the top of the cylinder is the top dead centre
- **Bottom dead centre(BDC):** Position of the piston when it is farthest from the top of the cylinder is the bottom dead centre
- **Stroke:** Travel of the piston from one dead centre to the other.
- **Stroke length:** The distance between the two dead centres
- **Swept volume:** Volume of the cylinder in between the two dead centres
- **Clearance volume:** Volume of the cylinder in between the top dead centre and the cylinder head

Swept Volume, Clearance Volume and Total Volume

Swept volume is the volume displaced by the piston in one stroke. It is also called **piston displacement volume**. It is denoted by V_s and can be calculated using the equation,

$$V_s = \text{area of piston} \times \text{length of stroke}$$

or

$$V_s = (\pi/4)d^2 L$$

Clearance volume is the volume left in the cylinder when the piston reaches the top dead centre position. It is denoted by V_c .

Total volume is the maximum volume of the cylinder. It is the sum of swept volume and the clearance volume and is denoted by V_t .

$$V_t = V_s + V_c$$

Compression Ratio and Clearance Ratio

Compression ratio is the ratio of maximum volume (total volume) to the minimum volume (clearance volume). It is denoted by r

$$r = \frac{V_t}{V_c} = \frac{V_s + V_c}{V_c}$$

Clearance ratio is the ratio of clearance volume to swept volume. It is denoted by c .

$$c = \frac{V_c}{V_s}$$

Mean Effective Pressure

Mean effective pressure (p_m) is the hypothetical average pressure, which if acted on the piston during the entire power stroke, will produce the same power output as produced during the actual cycle. According to the definition, we have

Work done/cycle = Force on piston $\times$ distance moved by piston

$$\begin{aligned} W &= (p_m \times \text{area of piston}) \times \text{length of stroke} \\ &= p_m \times V_s \end{aligned}$$

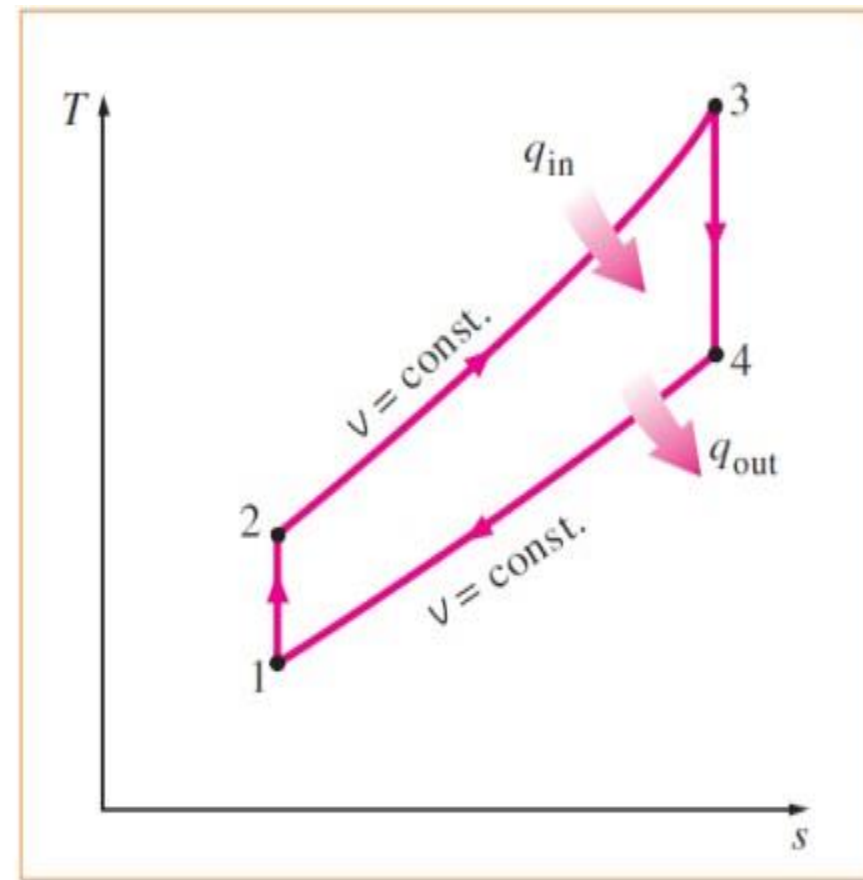
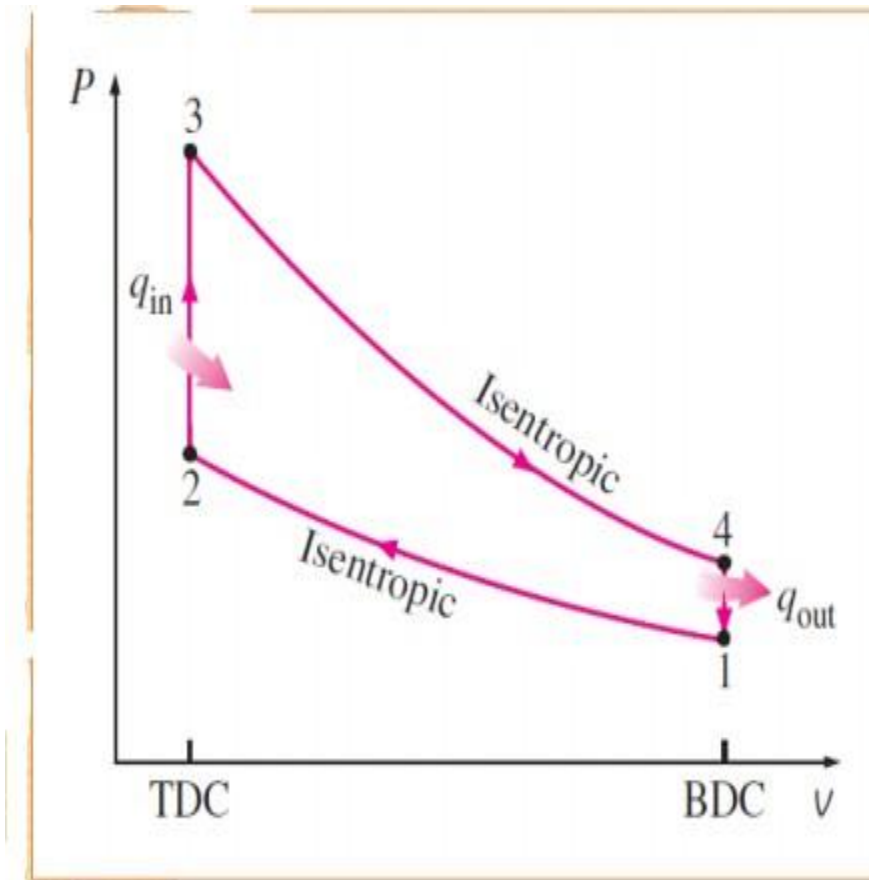
or

$$p_m = \frac{W}{V_s}$$

OTTO CYCLE

- *The Otto cycle is the ideal cycle for spark-ignition reciprocating(Petrol) engines. It is named after Nikolaus A. Otto, who built a successful four stroke engine in 1876 in Germany using the cycle proposed by Frenchman Beau de Rochas in 1862*
- **Otto cycle forms the basis for the petrol engine**
- **The cycle is:**
 - 1-2: Isentropic compression**
 - 2-3: Heat addition at constant volume**
 - 3-4: Isentropic Expansion**
 - 4-1:Heat rejection at constant volume**

p-V & T-S diagram



- Process 1-2, Isentropic compression
- Process 2-3, Constant volume heat addition
- Process 3-4, Isentropic expansion(Power Stroke)
- Process 4-1, Constant volume heat rejection

- Compression Ratio, $r = V_1/V_2$; also Expansion Ratio, $r = V_4/V_3$
- **Heat Supplied, $Q_{in} = mc_v(T_3-T_2)$**
- **Heat Rejected, $Q_{out} = mc_v(T_4-T_1)$**
- **Work Done, $W_{net} = Q_{in} - Q_{out}$**
- **Thermal Efficiency, $\eta = 1 - (Q_{out}/Q_{in})$**

$$= 1 - [mc_v(T_4-T_1) / mc_v(T_3-T_2)]$$

$$= 1 - [1/r^{(\gamma-1)}]$$

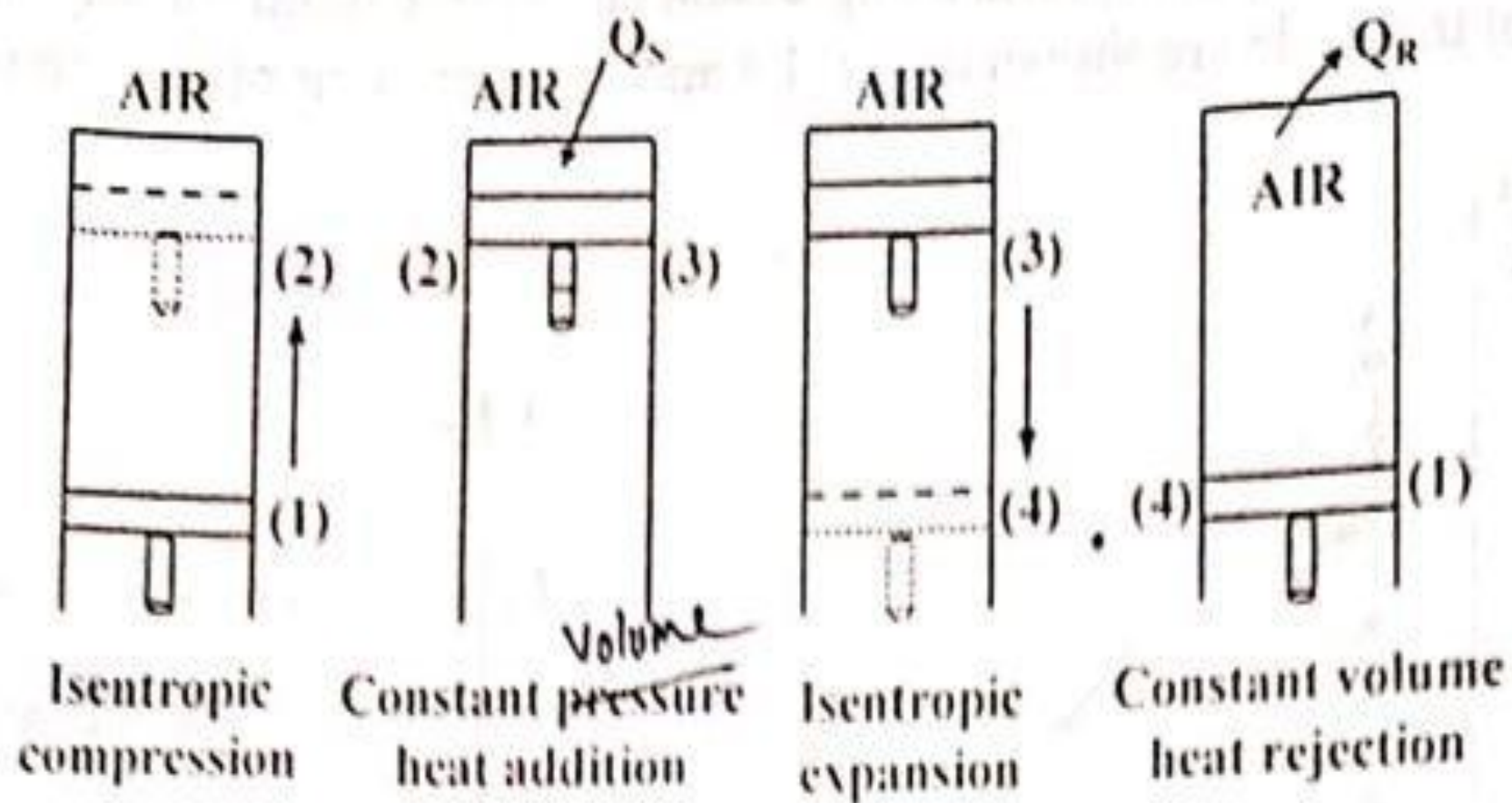


Fig. Process in Otto cycle

Heat supplied during constant volume process 2-3,

$$Q_S = C_v (T_3 - T_2)$$

Heat rejected during constant volume process 4-1,

$$Q_R = C_v (T_4 - T_1)$$

Air standard efficiency,

$$\eta = \frac{W}{Q_S} = \frac{Q_S - Q_R}{Q_S} = 1 - \frac{Q_R}{Q_S}$$

$$= 1 - \frac{C_v (T_4 - T_1)}{C_v (T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

(1)

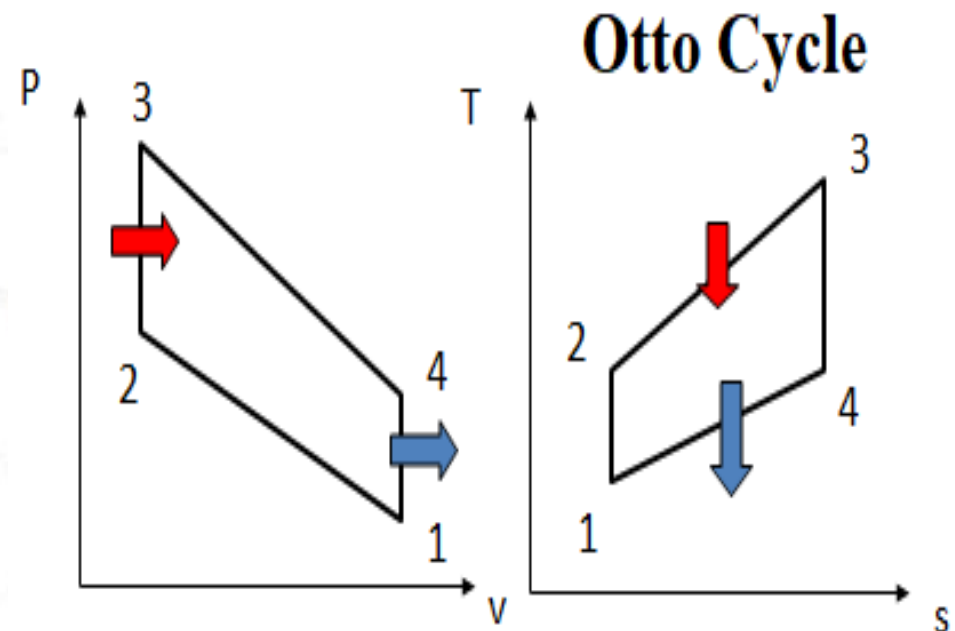
For the isentropic process 1-2 and 3-4, we have

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1} = r^{\gamma-1}$$

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3} \right)^{\gamma-1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1} = r^{\gamma-1}$$

(2)

$$\text{and } \frac{T_2}{T_3} = \frac{T_1}{T_4} r^{\gamma-1}$$



Substituting in Eq. (1), we get

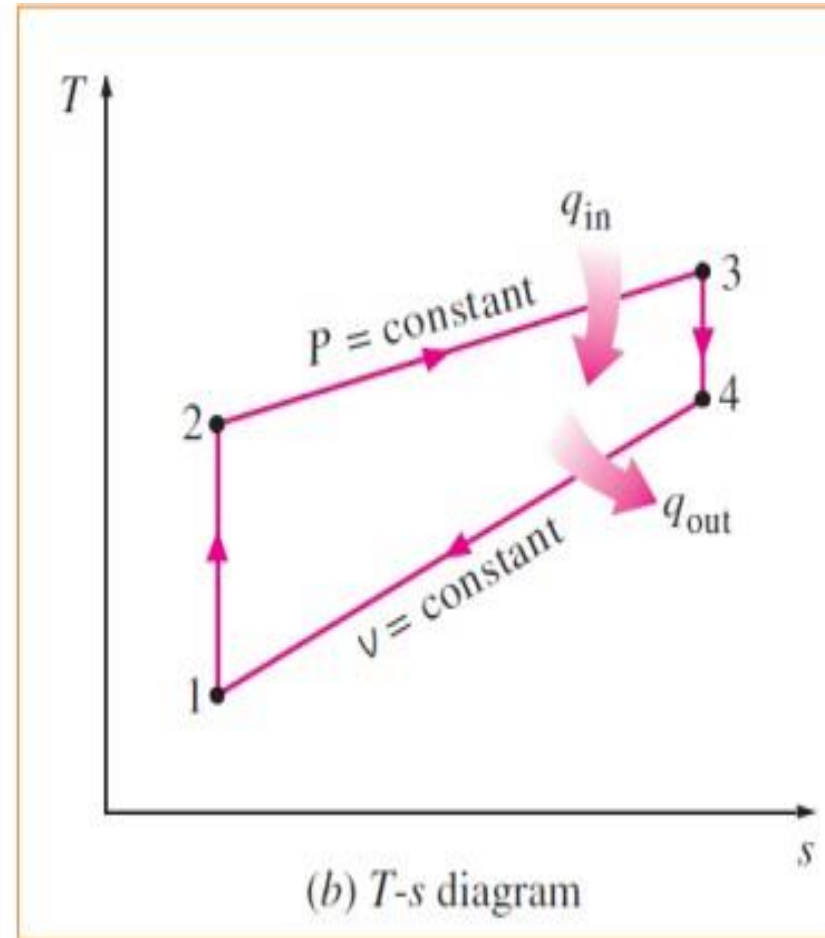
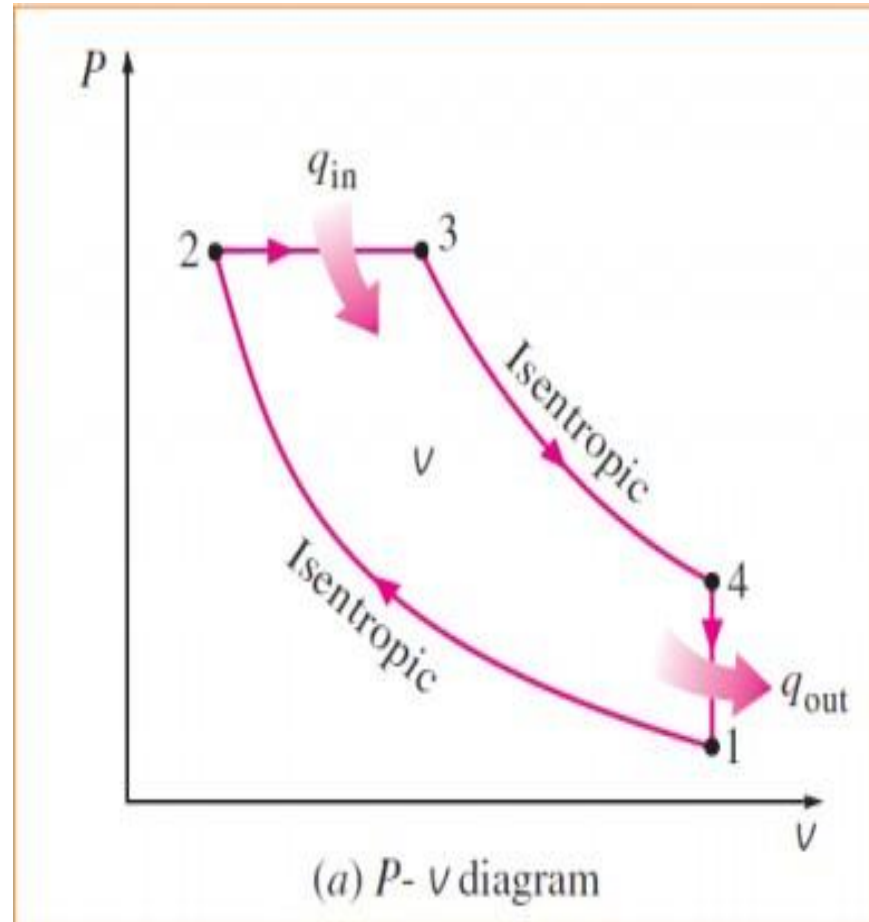
$$\eta = 1 - \frac{T_4 - T_1}{T_4 r^{\gamma-1} - T_1 r^{\gamma-1}} = 1 - \frac{T_4 - T_1}{(T_4 - T_1) r^{\gamma-1}}$$

$$\eta = 1 - \frac{1}{r^{\gamma-1}}$$

DIESEL CYCLE

- *The Diesel cycle is the ideal cycle for Compression Ignition(CI) reciprocating engines. The CI engine, first proposed by Rudolph Diesel in the 1890s*
- **Diesel cycle forms the basis for the diesel engine**
- **The cycle is:**
 - 1-2: Isentropic compression**
 - 2-3: Heat addition at constant pressure**
 - 3-4: Isentropic Expansion**
 - 4-1: Heat rejection at constant volume**

p-V & T-S diagram



- Process 1-2, Isentropic compression
- Process 2-3, Constant pressure heat addition
- Process 3-4, Isentropic expansion(Power Stroke)
- Process 4-1, Constant volume heat rejection

- **Compression Ratio, $r = V_1/V_2$**

- **Expansion Ratio, $r_1 = V_4/V_3$**

- **Cutoff Ratio, $\rho = V_3/V_2$**

- **$r_1 = r/\rho$**

- **Heat Supplied, $Q_{in} = mc_p(T_3-T_2)$**

- **Heat Rejected, $Q_{out} = mc_v(T_4-T_1)$**

- **Work Done, $W_{net} = Q_{in} - Q_{out}$**

- Thermal Efficiency, $\eta = 1 - (Q_{\text{out}}/Q_{\text{in}})$
 $= 1 - [mc_v(T_4 - T_1) / mc_p(T_3 - T_2)]$

$$\eta = 1 - \frac{1}{r^{\gamma-1}} \left(\frac{1}{\gamma} \right) \left[\frac{\rho^{\gamma} - 1}{\rho - 1} \right]$$

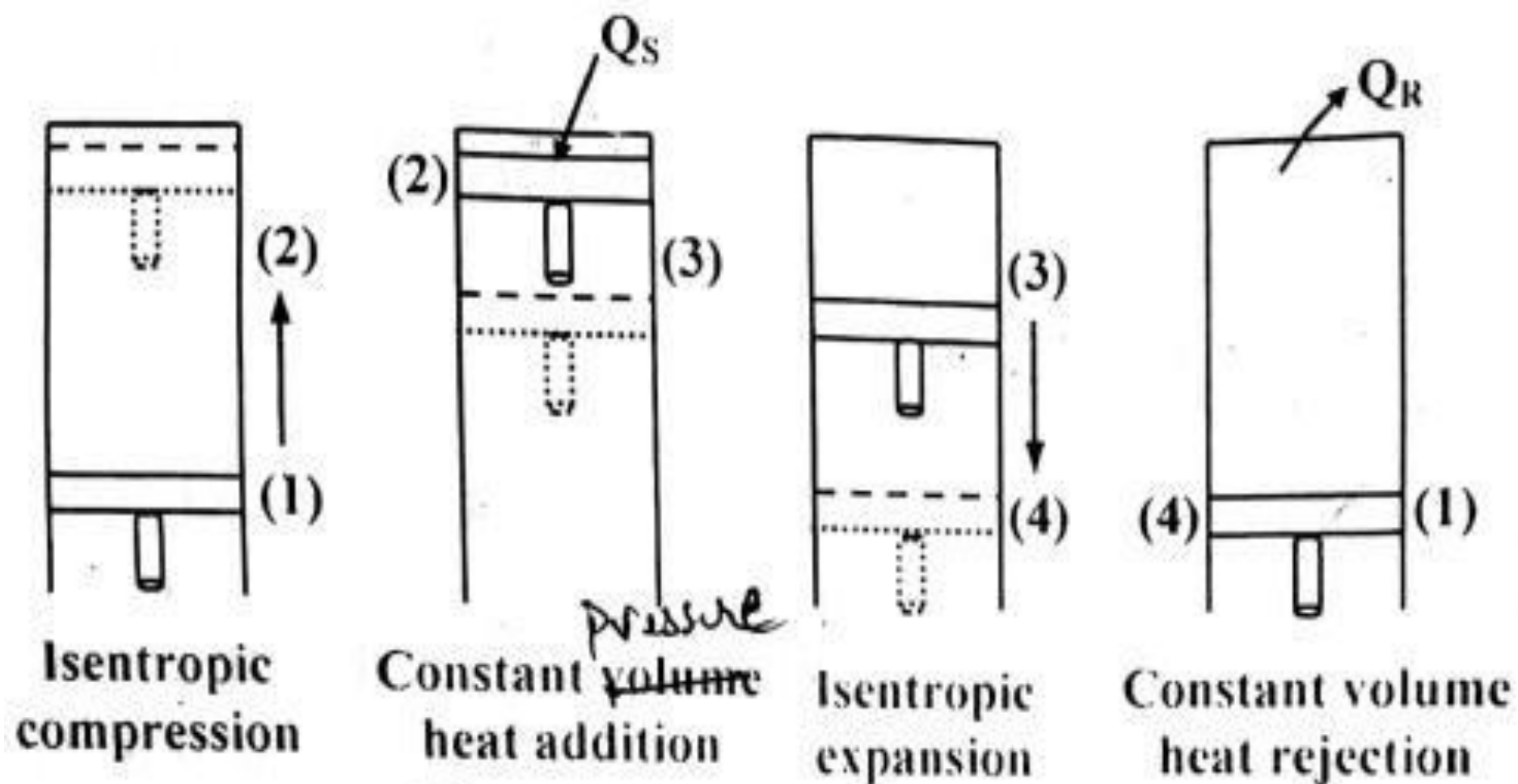


Fig. Process in Diesel cycle

Heat supplied during the constant pressure process 2-3,

$$Q_S = C_p (T_3 - T_2)$$

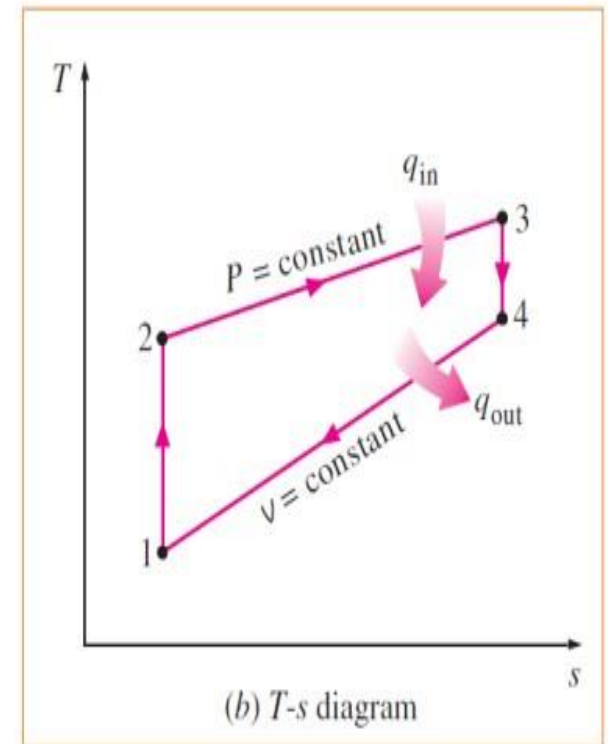
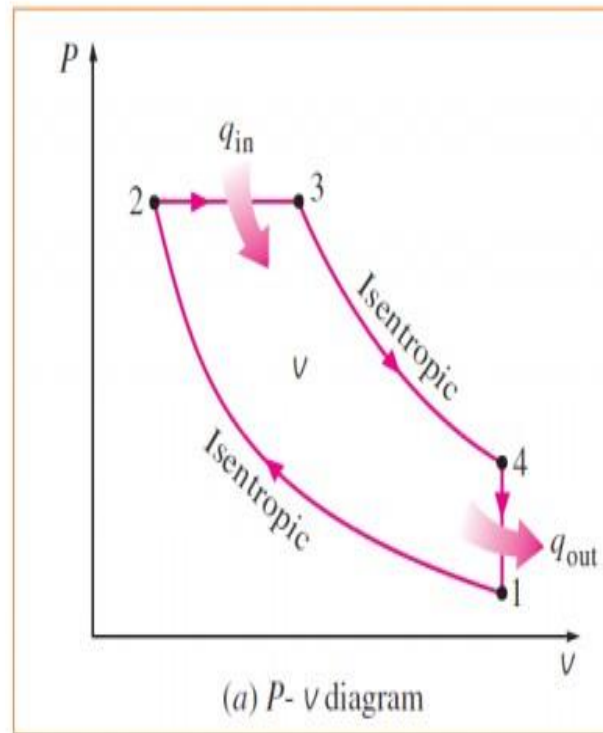
Heat rejected during the constant volume process 4-1,

$$Q_R = C_v (T_4 - T_1)$$

Air standard efficiency,

$$\eta = \frac{W}{Q_S} = \frac{Q_S - Q_R}{Q_S} = 1 - \frac{Q_R}{Q_S}$$

$$= 1 - \frac{C_v (T_4 - T_1)}{C_p (T_3 - T_2)} = 1 - \frac{1}{\gamma} \frac{(T_4 - T_1)}{(T_3 - T_2)} \quad \textcircled{1}$$



The efficiency of the Diesel cycle is usually expressed in terms of the following volume ratios.

i) Compression ratio (r)

$$r = \frac{V_1}{V_2} = \frac{\text{Volume before compression}}{\text{Volume after compression}}$$

ii) Expansion ratio (r_e)

$$r_e = \frac{V_4}{V_3} = \frac{\text{Volume after expansion}}{\text{Volume before expansion}}$$

iii) Cut off ratio (ρ)

$$\rho = \frac{V_3}{V_2} = \frac{\text{Volume after heat supply}}{\text{Volume before heat supply}}$$

A relation between the above three ratios can be obtained as

$$\begin{aligned} r &= \frac{V_1}{V_2} = \frac{V_1}{V_3} \times \frac{V_3}{V_2} = \frac{V_4}{V_3} \times \frac{V_3}{V_2} \quad (\because V_1 = V_4) \\ &= r_e \rho \end{aligned} \quad (2)$$

For the isentropic process 1-2,

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1} = r^{\gamma-1}$$

or

$$T_2 = T_1 r^{\gamma-1}$$

For the constant pressure process 2-3,

$$\frac{T_3}{T_2} = \frac{V_3}{V_2} = \rho$$

or

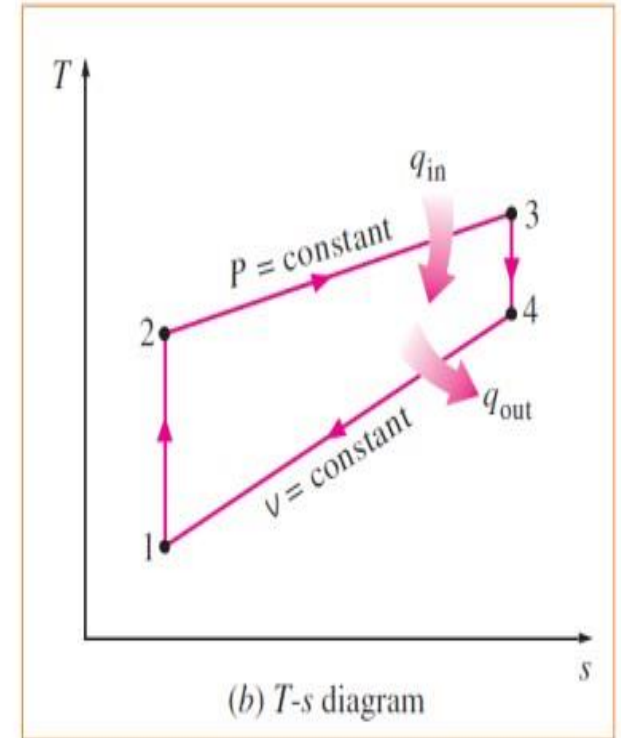
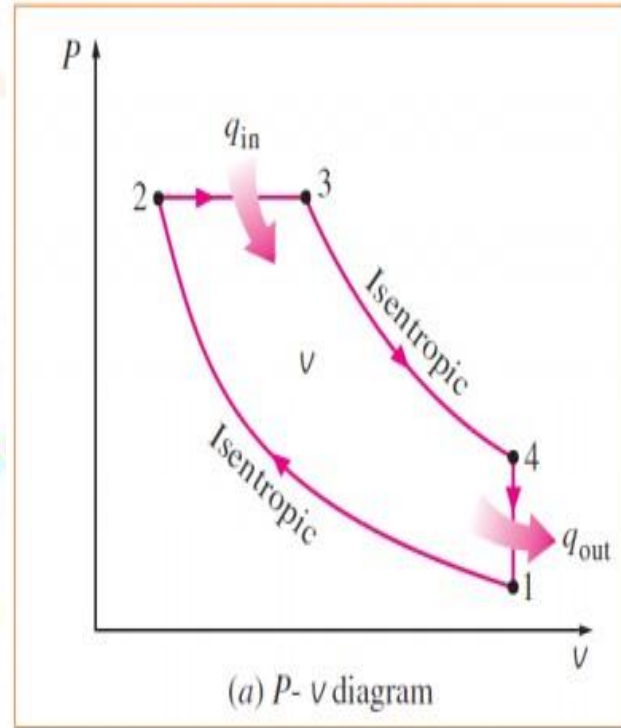
$$T_3 = \rho T_2 = \rho r^{\gamma-1} T_1$$

For isentropic process 3-4,

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4} \right)^{\gamma-1} = \left(\frac{V_3}{V_1} \right)^{\gamma-1} = \left(\frac{V_3}{V_2} \times \frac{V_2}{V_1} \right)^{\gamma-1} = \left(\frac{\rho}{r} \right)^{\gamma-1}$$

or

$$T_4 = \left(\frac{\rho}{r} \right)^{\gamma-1} T_3$$



$$= \left(\frac{\rho}{r} \right)^{\gamma-1} \rho r^{\gamma-1} T_1$$

$$= \rho^\gamma T_1$$

Substituting the values of T_2 , T_3 and T_4 in eqn. (1)
we get

$$\eta = 1 - \frac{1}{\gamma} \cdot \frac{(\rho^\gamma T_1 - T_1)}{(\rho r^{\gamma-1} T_1 - r^{\gamma-1} T_1)} = 1 - \frac{1}{\gamma} \frac{(\rho^\gamma - 1)}{(\rho r^{\gamma-1} - r^{\gamma-1})}$$

$$\eta = 1 - \frac{1}{r^{\gamma-1}} \left[\frac{\rho^\gamma - 1}{\gamma(\rho - 1)} \right]$$

An engine working on the Otto cycle has an efficiency of 56 %. It rejects heat at the rate of 544 kJ per kg of air. The pressure and temperature of air at the beginning of compression are 0.1 MPa and 60°C respectively. Compute,

- (i) The compression ratio of the engine
- (ii) The work done per kg of air
- (iii) The pressure and temperature at the end of compression and
- (iv) The maximum pressure in the cycle.

Assume suitable values for C_p and C_v .

Solution:

Given:

$$\eta = 56 \% Q_r = 544 \text{ kJ/kg,}$$
$$p_1 = 0.1 \text{ MPa} = 0.1 \times 10^6 \text{ N/m}^2$$
$$T_1 = 60^\circ\text{C} = 60 + 273 = 333 \text{ K}$$